

Operator Discovery Report: restart_controller_stagnation_escape_balanced

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically trigger small, bounded-restart growth under stagnation and enable perturbation-based restarts to escape 2-opt traps without excessive randomization across instance classes.
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:6:1.
- Surviving candidate: False.
- Problem class where it helps: two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: a0d77e9aa0626d3a.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Runtime inflation is high at 0.707399.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 220.714771, distance-eval delta 5781.714286.
- cheapest_insertion_2opt gap delta -0.002859, runtime delta 1023.507586, distance-eval delta 4670.285714.
- random_restart_2opt gap delta 0.0, runtime delta -185.635371, distance-eval delta -11541.714286.
- sparse_three_opt gap delta 0.034327, runtime delta 24.968514, distance-eval delta -7.857143.
- clustered_local_search gap delta -0.003587, runtime delta -112.344529, distance-eval delta -6283.142857.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.110431 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003304.
- Runtime inflation: 0.707399.
- Distance-evaluation inflation: 0.770256.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003304.
- fixed_restart_budget mean gap 0.118013 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.117688 and delta vs full operator -0.000325.